Wash derivation evidence

Why Wash displaces its ground in OKLab

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1 Purpose

quiet-core defines Wash as a displacement of the local ground in OKLab, toward the hue of a chromatic family, by a fixed distance. The line tier preserves the ground's lightness.

This report records the evidence for that definition: the alternatives that were rejected, the gamut limit on the line distance, and the reason Field is the canonical ground.

2 Requirements

A wash marks a region, such as a diff line or a diagnostic line, with the meaning of one chromatic family. The line tier must:

1. carry the family's hue visibly;
2. keep every foreground that is readable on Field readable on the wash;
3. give all four families the same perceptual strength;
4. remain inside sRGB without clipping.

In the tables below, *lowest* is the lowest contrast among the foregrounds that must reach 4.5:1: Muted and the four chromatic anchors. Ink, at 7:1, is never the binding constraint here.

3 Rejected: interpolation toward the anchor

Selection is derived by interpolating from Structure toward Mineral in OKLab. The same operation from Field toward each family anchor was evaluated first.

Table 1: Lowest 4.5-class contrast on Field interpolated toward each family.

Amount	quiet-day lowest	quiet-night lowest
0.04	4.36 Mineral on Rust mix	4.93 Rust on Ochre mix
0.06	4.25 Mineral on Rust mix	4.80 Rust on Ochre mix
0.08	4.14 Mineral on Rust mix	4.66 Rust on Ochre mix
0.10	4.03 Mineral on Rust mix	4.53 Rust on Ochre mix
0.15	3.77 Mineral on Rust mix	4.18 Rust on Ochre mix
0.20	3.52 Mineral on Rust mix	3.84 Rust on Ochre mix

The chromatic anchors are readable foregrounds, so they are far from Field in lightness. Interpolating toward them mostly darkens quiet-day rather than tinting it. quiet-day falls below 4.5:1 at every amount evaluated.

quiet-night tolerates small amounts, but one definition must serve both adaptations. Interpolation fails requirement 2.

4 Rejected: fixed absolute chroma

The second alternative kept Field's lightness and set a fixed absolute OKLCH chroma at each family's hue. The table shows the resulting distance from Field.

quiet-day Field is warm: its own hue lies near Ochre. At a fixed chroma, Ochre barely moves away from Field while Mineral moves several times as far. Fixed chroma fails requirement 3.

Table 2: Distance from Field at a fixed chroma of 0.025.

Family	quiet-day ΔE	quiet-night ΔE
Mineral	0.037	0.024
Moss	0.022	0.020
Ochre	0.011	0.023
Rust	0.018	0.027

5 Chosen: displacement by a fixed distance

The chosen rule adds a fixed-length vector in OKLab's chromatic plane, pointing along the family's hue, to the ground. Lightness is unchanged in the line tier, so distance from Field is identical for every family by construction.

Table 3: Canonical line washes over Field.

Family	quiet-day ΔE	quiet-day lowest	quiet-night ΔE	quiet-night lowest
Mineral	0.020	4.62 Mineral	0.020	5.16 Rust
Moss	0.020	4.63 Mineral	0.020	5.16 Rust
Ochre	0.020	4.59 Mineral	0.020	5.19 Rust
Rust	0.020	4.57 Mineral	0.020	5.21 Rust

Every family moves the same distance, and the lowest 4.5-class contrast stays above the floor in both adaptations. Requirements 1 to 3 hold.

6 Gamut limit

A displacement can leave sRGB. quiet-core treats that as a derivation failure rather than a color to clip. The table gives, for each family, the largest line distance over Field that remains in gamut.

Table 4: Largest in-gamut line distance over Field.

Family	quiet-day ceiling	quiet-night ceiling
Mineral	0.0660	0.0381
Moss	0.1077	0.0708
Ochre	0.0394	0.0449
Rust	0.0277	0.0769

One line distance is shared by all families, so it is bounded by the smallest ceiling in either adaptation. quiet-day Rust sets that bound. The chosen distance of 0.020 leaves headroom below it. The test suite fails if any ceiling falls to or below the line distance.

7 Canonical ground

A wash depends on its ground. The table evaluates every family's line wash over each architectural background.

Over Field, both adaptations pass. Over quiet-day Surface and quiet-night Structure, the lowest foreground falls just below 4.5:1.

Field is therefore the canonical ground, and the contract evaluates washes over Field. A port that places a wash on another ground must derive it over that ground and verify the foregrounds it actually draws there.

8 Emphasis tier

The emphasis tier marks a span inside a line wash and is intended for Ink. It moves lightness away from the ground so that it can carry more chroma within gamut.

quiet-core requires only Ink at 7:1 on each emphasis wash. Those results appear in the conformance report.

9 Independence

The canonical wash values asserted by the `internal/core` tests were computed by an independent implementation of the specified rule, not by the reference implementation. The tables here document the reference implementation; the fixtures check it.

Table 5: Lowest 4.5-class contrast over line washes on each ground.

Ground	quiet-day lowest	quiet-night lowest
Field	4.57 Mineral on Rust wash	5.16 Rust on Moss wash
Surface	4.48 Mineral on Rust wash	4.55 Rust on Moss wash
Structure	4.52 Mineral on Rust wash	4.49 Rust on Moss wash